Real-Time Upper Body Posture Classification

via MediaPipe and a Hybrid ML–Geometric Approach

Signal, Image and Video Project Report · A.Y. 2025–2026

Francesca Loffredo & Kevin Shimaj

Abstract. This report presents a system for real-time upper body posture classification using skeletal landmarks extracted by MediaPipe Pose from a standard webcam feed. Five posture classes are recognised: upright (TUP), forward lean (TLF), backward lean (TLB), lateral-left (TLL), and lateral-right (TLR). A *hybrid* strategy is adopted: a machine-learning model — specifically a Random Forest trained on three subject-normalised geometric features — classifies TUP, TLF, and TLB, while explicit geometric rules handle TLL and TLR. The best model achieves **96.2% accuracy** and a **macro F1 of 0.963** on a held-out group split, with 98.9% recall on the minority class TLB. A subject-specific calibration phase and temporal smoothing ensure robust real-time performance.

1. Introduction

Poor posture during prolonged seated activity is associated with musculoskeletal discomfort and long-term health risks. Automated, real-time feedback systems can support users in correcting their posture without requiring wearable sensors or specialised hardware. This project develops such a system relying solely on a standard RGB camera (an iPhone used as a webcam) and the open-source MediaPipe Pose library for skeletal tracking.

The system recognises five upper-body posture classes (Table 1). Three of them — TUP, TLF, and TLB — describe sagittal-plane and vertical postures and are handled by a supervised machine-learning classifier. The remaining two — TLL and TLR — describe lateral tilts and are detected through explicit geometric rules, motivated by the symmetric nature of lateral deviation and the high risk of confusion with other classes when using learned boundaries alone.

Table 1: *Posture classes and detection strategy.*

Class	Full Name	Description	Detection
TUP	Trunk Upright	Neutral, upright seated posture	ML
TLF	Trunk Lean Forward	Forward inclination of head and trunk	ML
TLB	Trunk Lean Backward	Backward inclination, reclined posture	ML
TLL	Trunk Lean Left	Lateral tilt toward the left side	Geometric
TLR	Trunk Lean Right	Lateral tilt toward the right side	Geometric

The overall pipeline is divided into three main phases: (i) an *offline* phase covering dataset construction, landmark selection, feature engineering, and model training; (ii) a *per-session calibration* phase that establishes a subject-specific baseline in the neutral TUP posture; and (iii) a *real-time classification loop* that continuously extracts features and emits posture labels with confidence scores.

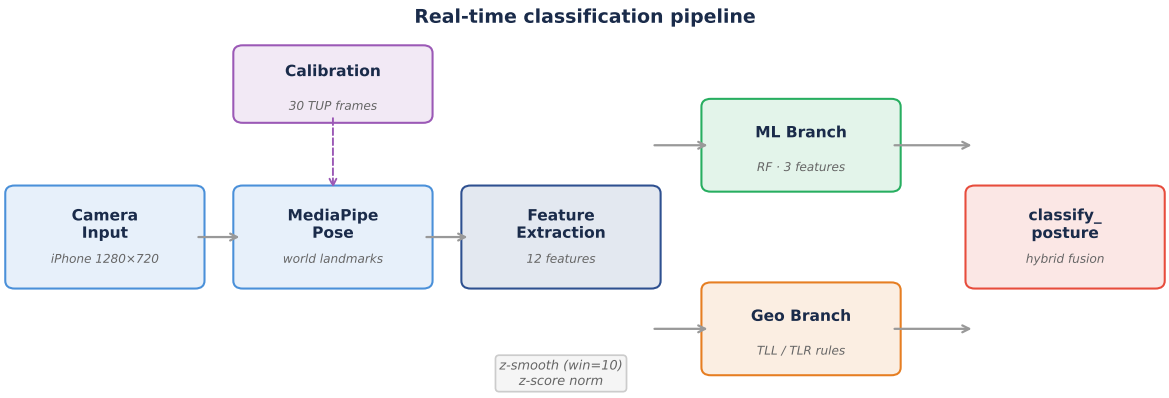

Figure 1: *Overview of the real-time classification pipeline. A calibration phase (dashed arrow) establishes the per-subject baseline before the main loop starts. Features are split between the ML branch (TUP/TLF/TLB) and the geometric branch (TLL/TLR), whose outputs are fused by* `classify_posture`.

2. Dataset

2.1 Original Dataset

The starting point for this work is the publicly available **MultiPosture** dataset (Carneros-Prado et al., 2024), released under open access on Zenodo (https://zenodo.org/records/14230872). It contains skeletal pose data extracted from video recordings of **13 participants** performing

various sitting postures in home environments, processed using the MediaPipe Pose Heavy model. The dataset includes 4,800 frames of 3D skeletal coordinates for 11 key body joints, each frame manually labelled for both upper and lower body posture.

Original coordinate system. The MultiPosture dataset stores landmark coordinates *normalised relative to the hip centre*: all positions are expressed with respect to a reference frame whose origin is placed at the midpoint between the two hip landmarks. In this system the hip centre is always at $(0, 0, 0)$, the y-axis points *upward*, and distances are metric (metres). As a result, upper-body landmarks such as the nose have a consistently *negative* y-coordinate (they lie above the origin in the world, which means smaller y values in the upward-positive convention).

2.2 Re-acquisition with iPhone Camera

The original MultiPosture data was collected with a frontal, approximately perpendicular camera setup. Our deployment scenario is different: the video is captured using an **iPhone mounted at roughly 30° downward angle**, positioned on a desk or monitor stand looking down at the seated user. This oblique viewpoint modifies how landmarks project onto the image plane and, more critically, affects the *depth* (z) estimates provided by MediaPipe, which uses monocular depth inference.

To ensure the training data matches the inference conditions, we re-processed the original recordings — or collected new samples under the same angular conditions — using `pose_world_landmarks` from MediaPipe Pose. The world-coordinate output normalises each frame so that the hip midpoint is approximately at the origin regardless of camera angle, making features computed in world coordinates intrinsically more robust to viewpoint variation than image-plane features. The validity of the world-coordinate system in our data was confirmed empirically: the median nose y-coordinate across all samples is -0.49 and the median hip y-coordinate is -0.002, consistent with the expected upward-positive convention.

2.3 Final Dataset Statistics

The raw dataset used for training contains **4,794 samples** $\times$ **102 columns**, including x, y, z coordinates and visibility scores for all 33 landmarks, plus subject identifier and posture label.

Table 2: *Class distribution in the full dataset.*

Class	Samples	% of Total	Pipeline
TLF	1,897	39.6%	ML
TUP	1,615	33.7%	ML
TLB	442	9.2%	ML (minority)
TLL	420	8.8%	Geometric
TLR	420	8.8%	Geometric
Total	**4,794**	**100%**	—

TLB is significantly under-represented relative to TLF and TUP. This imbalance is addressed at training time via class-weight rebalancing.

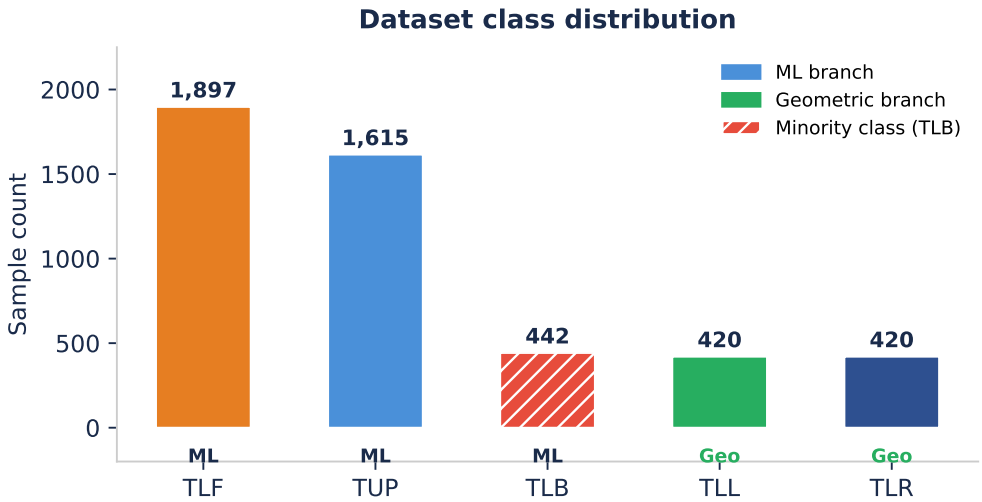

Figure 2: *Class distribution across the 4,794 samples. TLB (hatched bar) represents only 9.2% of the ML training set and is the minority class targeted by class-weight rebalancing. TLL and TLR are handled entirely by geometric rules and are excluded from ML training.*

3. Landmark Selection

From the 33 available MediaPipe landmarks, nine were selected as sufficient and robust for upper-body posture characterisation (Table 3). This selection reduces the dataset from 102 to **29 columns**, eliminating lower-body landmarks that are uninformative for the target classes and often occluded or unreliable in seated settings.

Table 3: *Selected landmarks and their role in the feature pipeline. (*) Hip landmarks are used only when confidence* ≥ 0.6.

Landmark	Axes	Role
Nose	x, y, z	Head position and neck angle reference
Left Eye	x, y, z	Head tilt (primary)
Right Eye	x, y, z	Head tilt (primary)
Left Ear	x, y, z	Head tilt (fallback) + ear asymmetry
Right Ear	x, y, z	Head tilt (fallback) + ear asymmetry
Left Shoulder	x, y, z	Shoulder slope, trunk inclination
Right Shoulder	x, y, z	Shoulder slope, trunk inclination
Left Hip*	x, y, z	Trunk sagittal angle (support feature)
Right Hip*	x, y, z	Trunk sagittal angle (support feature)

Confidence thresholds are applied per landmark type: `HEAD_Z_CONF_THR = 0.6`, `SHOULDER_CONF_THR = 0.6`, `HIP_CONF_THR = 0.6`. A higher threshold (`EAR_Z_CONF_THR = 0.7`) is used for ear depth, which is more sensitive to head rotation. Landmarks below threshold are flagged as unavailable and trigger fallback paths in the feature computation.

4. Midpoint Construction

Three virtual reference points are derived from the selected landmarks before any feature is computed: **mid_shoulder**, **mid_hip**, and **head_center**.

4.1 Bilateral Midpoints

mid_shoulder and **mid_hip** are computed as the arithmetic mean of the corresponding bilateral landmark pairs:

$$\mathbf{m}_{\text{shoulder}} = \frac{\mathbf{p}_{\text{L-shoulder}} + \mathbf{p}_{\text{R-shoulder}}}{2}, \qquad \mathbf{m}_{\text{hip}} = \frac{\mathbf{p}_{\text{L-hip}} + \mathbf{p}_{\text{R-hip}}}{2} \tag{1}$$

These midpoints provide stable, symmetric reference axes for trunk-level features and reduce sensitivity to minor left–right asymmetries. Each midpoint is accompanied by an *availability flag*, set to `True` only when both constituent landmarks exceed the confidence threshold. The combined visibility of a midpoint is defined as the geometric mean of the two individual visibilities.

4.2 Head Center

head_center is a single representative point for the head, computed with a priority-based fallback strategy to maximise availability across different head orientations and occlusion conditions:

1. **Eyes available (primary):** if both the left and right eye landmarks are valid and their combined confidence exceeds `HEAD_Z_CONF_THR = 0.6`, head_center is set to the midpoint of the two eyes. This is the most accurate estimate because the eyes are close to the head's geometric centre and are reliably detected in frontal views.
2. **Ears available (fallback):** if the eyes are not reliable but both ears are valid with confidence $\geq$ `EAR_Z_CONF_THR = 0.7`, head_center is set to the midpoint of the two ears. The ear threshold is intentionally higher because ear depth estimates from monocular MediaPipe are noisier, especially under the 30° camera angle used in our setup.
3. **Nose only (last resort):** if neither bilateral estimate is available, head_center falls back to the nose landmark alone.

$$\mathbf{h} = \begin{cases} \frac{1}{2}(\mathbf{p}_{\text{L-eye}} + \mathbf{p}_{\text{R-eye}}) & \text{if eyes reliable} \\ \frac{1}{2}(\mathbf{p}_{\text{L-ear}} + \mathbf{p}_{\text{R-ear}}) & \text{else if ears reliable} \\ \mathbf{p}_{\text{nose}} & \text{otherwise} \end{cases} \tag{2}$$

head_center is used instead of the raw nose position in all features that require a stable head location (`head_lateral_ratio`, `head_y_ratio`, `head_neck_vertical_angle`), making those features more robust to changes in head rotation.

Summary of derived reference points.
mid_shoulder — always required; used in all primary and support features.
mid_hip — used only when both hips are reliable (confidence ≥ 0.6); enables trunk sagittal features.
head_center — always available via fallback; replaces raw nose in head-position features.

5. Feature Engineering

Twelve geometric features are computed from the selected landmarks, divided into *primary features* (head and shoulder geometry, always available) and *support features* (incorporating hip information when reliable). A final subset of three features is selected for the ML model.

5.1 Primary Features

Primary features capture upper body posture without relying on the pelvis.

Table 4: *Primary features derived from head and shoulder landmarks.*

Feature	Description
`head_tilt`	Roll angle of the head from the inter-eye or inter-ear axis; wrapped to $[-90, 90)$.
`shoulder_slope`	Inclination of the shoulder line w.r.t. the horizontal; positive = right shoulder higher.
`head_lateral_ratio`	Normalised horizontal offset of the nose from mid_shoulder, divided by shoulder width.
`head_y_ratio`	Vertical position of the nose relative to mid_shoulder, divided by shoulder width. Negative in TUP (head above shoulders in world coords).
`head_neck_vertical_angle`	Angle between the nose-to-mid_shoulder vector and the vertical axis.
`head_shoulder_alignment`	Signed horizontal difference between nose x and mid_shoulder x, normalised.
`upper_body_inclination`	Sagittal inclination of the upper body estimated from the shoulder-to-nose vector.
`head_trunk_diff`	Difference between head_neck_vertical_angle and upper_body_inclination; isolates neck flexion from trunk lean.

5.2 Support Features

Support features incorporate pelvis-based information and are computed only when `mid_hip` is available; otherwise they are set to `NaN` and imputed.

Table 5: *Support features derived from hip and bilateral ear landmarks.*

Feature	Description
`trunk_forward_backward_angle`	Sagittal inclination angle of the trunk vector ($\mathbf{m}_{\text{shoulder}} - \mathbf{m}_{\text{hip}}$) w.r.t. the vertical. Primary discriminator for TLF vs. TLB.
`shoulder_z_asym`	Depth asymmetry between the two shoulders ($z_{\text{L}} - z_{\text{R}}$). Captures axial body rotation.
`hip_shoulder_lateral`	Lateral offset of mid_shoulder from mid_hip, normalised by shoulder width.
`ear_height_asym`	Vertical height difference between left and right ears, normalised.

5.3 Final ML Feature Set and Normalisation

After evaluating all twelve features for inter-class separability and cross-subject stability, three features are selected for the ML model:

- `head_y_ratio` — the single most discriminative feature for TUP vs. TLF: in TUP the head sits well above mid_shoulder (negative value in world coords); in TLF the head drops forward and the ratio increases.
- `head_neck_vertical_angle` — captures the forward/backward direction of head tilt regardless of shoulder position.

- `trunk_forward_backward_angle` — the sagittal trunk angle; key for separating TLF from TLB.

Because subjects differ in body proportions and resting posture, a **per-subject z-score normalisation** is applied. Each feature is standardised using the mean and standard deviation computed over the subject's own TUP frames. A minimum standard-deviation floor is enforced to prevent over-amplification: $\sigma_{\text{floor}} = 0.05$ for `head_y_ratio` and $\sigma_{\text{floor}} = 4.0$ for both angular features. This normalisation is replicated at inference time during the calibration phase.

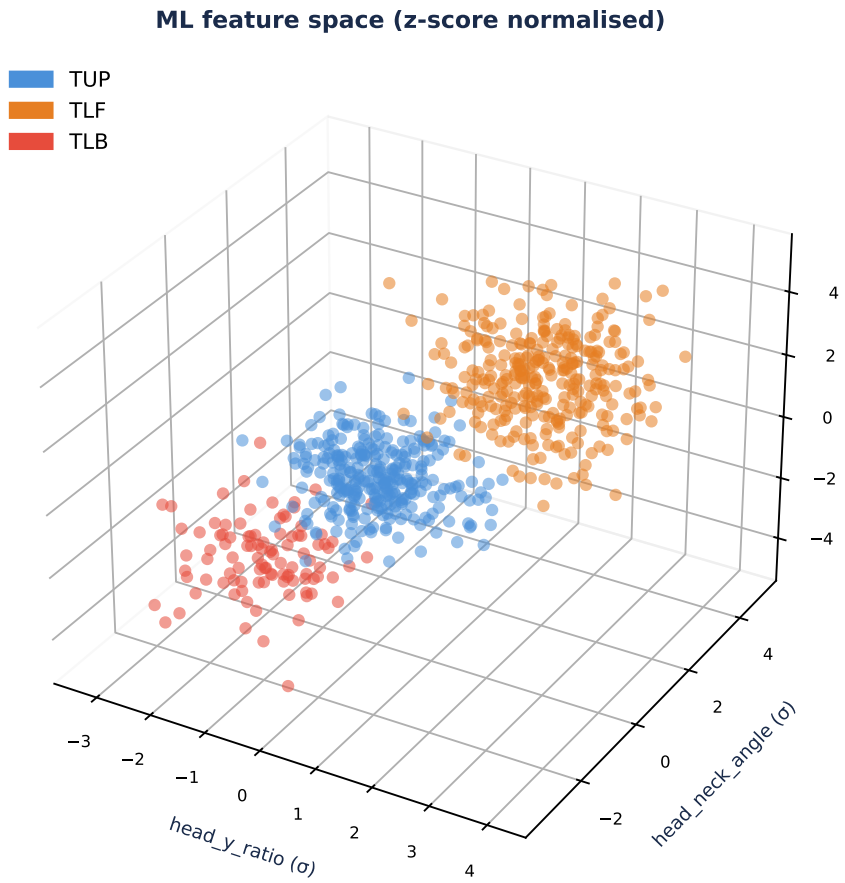

Figure 3: *3D scatter of the three ML features after per-subject z-score normalisation. TUP clusters near the origin; TLF shifts toward higher* `head_y_ratio` *and positive trunk angle (forward lean); TLB is displaced in the opposite direction (backward lean). The three classes are well-separated in this normalised space, justifying the use of a simple 3-feature classifier.*

6. Classification Architecture

The system adopts a **hybrid classification strategy** that assigns the five posture classes to two different detection mechanisms: a machine-learning model for TUP, TLF, and TLB, and explicit geometric rules for TLL and TLR. This split is not arbitrary — it follows directly from the physical constraints of the acquisition setup and from the reliability of the features that MediaPipe can provide under those constraints.

Why ML for TUP, TLF, TLB?

The three sagittal-plane postures differ primarily in how far the trunk and head lean *forward or backward* with respect to the hips. The key discriminating information lies in the **y-coordinates** of the landmarks (vertical position) and in the **z-coordinates** (depth, i.e. front-to-back distance). In our setup the camera is positioned *in front of the subject*, looking slightly downward at ≈ 30. From this frontal viewpoint, forward and backward body lean is directly visible as a change in

the relative height and depth of the head and shoulders with respect to the hips — exactly what features such as `head_y_ratio` and `trunk_forward_backward_angle` capture.

However, the boundary between TUP, TLF, and TLB is *not* a simple threshold: it depends on individual body proportions, habitual resting posture, and the non-linear way lean interacts with the camera projection. A learned model can capture these non-linearities from data, while a fixed threshold would need to be tuned per subject. The per-subject z-score normalisation described in Section 5 brings all subjects onto a common scale, making the learned boundary generalisable.

Why geometric rules for TLL and TLR?

Lateral postures (leaning left or right) are fundamentally different from a measurement perspective. The distinguishing signal is a **horizontal displacement** and a **tilt of the shoulder line and head**, both of which manifest primarily in the *x-coordinates* (lateral axis) and are clearly visible from any frontal camera angle. In this sense, TLL and TLR are *easier* to detect geometrically than the sagittal postures.

The deeper problem is that including TLL and TLR in the ML training set would *require the model to rely on the z-axis* to distinguish them from TLF and TLB, since in the feature space all five classes partly overlap in the y-plane. However, the z-axis (depth) estimated by MediaPipe is inherently the least reliable coordinate: it is inferred from a single monocular camera using a learned prior, and its accuracy degrades significantly when the camera is not perfectly perpendicular to the subject's frontal plane — exactly the case in our 30 downward-tilted setup. Training a classifier on a z-dependent boundary would therefore produce a model that is fragile to small changes in camera position, distance, or posture.

Geometric rules based on `shoulder_slope`, `ear_height_asym`, and `head_lateral_ratio` use only *x- and y-coordinates*, which are stable and well-estimated from a frontal view. These rules are also **interpretable** and **symmetric by design**: a large positive shoulder slope signals TLR, a large negative slope signals TLL, with a per-subject baseline normalisation that removes individual asymmetries. The result is a detector that is more robust to camera variability than any z-dependent learned boundary would be.

> **In summary:** TUP, TLF, TLB are separated by sagittal cues (depth + height) that require a learned model to handle individual variability. TLL and TLR are separated by lateral cues (x-axis only) that are reliably captured by simple geometric thresholds, avoiding any dependence on the noisy monocular depth estimate.

6.1 ML Branch: TUP, TLF, TLB

The ML branch receives the three z-score-normalised features and outputs a probability distribution over $\{\text{TUP}, \text{TLF}, \text{TLB}\}$. The predicted label is the argmax of this distribution. Class weights are set to account for the minority status of TLB.

6.2 Geometric Branch: TLL, TLR

Four geometric features are evaluated against calibrated thresholds to detect lateral tilt:

- **shoulder_slope** — unilateral shoulder elevation signals lateral lean.
- **shoulder_z_asym** — depth difference between shoulders reveals the axial rotation that accompanies lateral tilt.
- **ear_height_asym** — direct measure of head lateral tilt.
- **head_lateral_ratio** — displacement of the head from the shoulder midline.

All quantities are expressed as deltas from the subject's calibrated TUP baseline (in units of per-subject standard deviation). A weighted combination forms a *lateral score*; if it exceeds a threshold the frame is classified as TLL or TLR, overriding the ML prediction.

6.3 Hybrid Fusion: `classify_posture`

The function `classify_posture` implements the final decision logic:

1. **Extreme lateral guard:** if the lateral score is very high, the label is immediately set to TLL or TLR regardless of ML output.
2. **TLB confirmation:** if ML predicts TLB and sagittal features confirm a backward lean, TLB is accepted; otherwise the label may be revised to TUP.
3. **Lateral-vs-sagittal arbitration:** if ML predicts TLF but the lateral score is moderate, the geometric score takes precedence to avoid misclassifying lateral postures as forward lean.
4. **TUP floor:** the TUP probability is boosted by a constant (`TUP_FLOOR = 1.0`) to reduce spurious detections in ambiguous frames.
5. **Hysteresis:** a label change requires a margin of 0.15 above the current label's score to avoid rapid oscillation.

7. Model Training and Results

7.1 Experimental Setup

The ML model is trained exclusively on the three ML-target classes (TUP, TLF, TLB), yielding 3,954 samples after filtering. The dataset is split **by subject** using `GroupShuffleSplit` to ensure that no subject appears in both training and test sets:

- **Training set:** 10 subjects — 3,114 samples.
- **Test set:** 3 subjects — 840 samples.

Several model families are evaluated within a `sklearn` Pipeline (StandardScaler + classifier). Class weights are varied systematically to improve recall on the minority class TLB.

7.2 Ranking Metric

Models are ranked by a composite *deploy_score* combining macro F1, TLB recall, and TLB F1, reflecting the operational requirement that TLB must not be missed even at a small cost in overall accuracy.

7.3 Results

Table 6: *Model comparison on the group-split test set (3 subjects, 840 samples).*

Model	Acc.	Macro F1	TLB Rec.	TLB F1	Deploy
Random Forest balanced	**96.2%**	**0.963**	**98.9%**	**0.972**	**0.978**
Random Forest TLBx3	96.4%	0.966	98.4%	0.977	—
SVM balanced	96.7%	0.966	99.5%	0.955	—
SVM TLBx3	95.5%	0.956	99.5%	0.962	—
Gradient Boosting	96.0%	0.962	97.9%	0.976	—
Logistic Regression balanced	95.2%	0.952	100%	0.941	—
MLP	94.5%	0.948	98.4%	0.964	—
KNN	93.7%	0.936	90.0%	0.922	—

Selected model: Random Forest (balanced class weights)

Features:	3 (head_y_ratio, head_neck_vertical_angle, trunk_forward_backward_angle)	Accuracy:	96.2%
		Macro F1:	0.963
		TLB Recall:	98.9%

The Random Forest with balanced class weights achieves the best deploy score. With only three features it reaches 96.2% accuracy, a macro F1 of 0.963, and 98.9% recall on TLB — the most critical class, as missing a backward lean would fail to alert the user. The model is serialised with `pickle` together with the label encoder and feature column list for deployment.

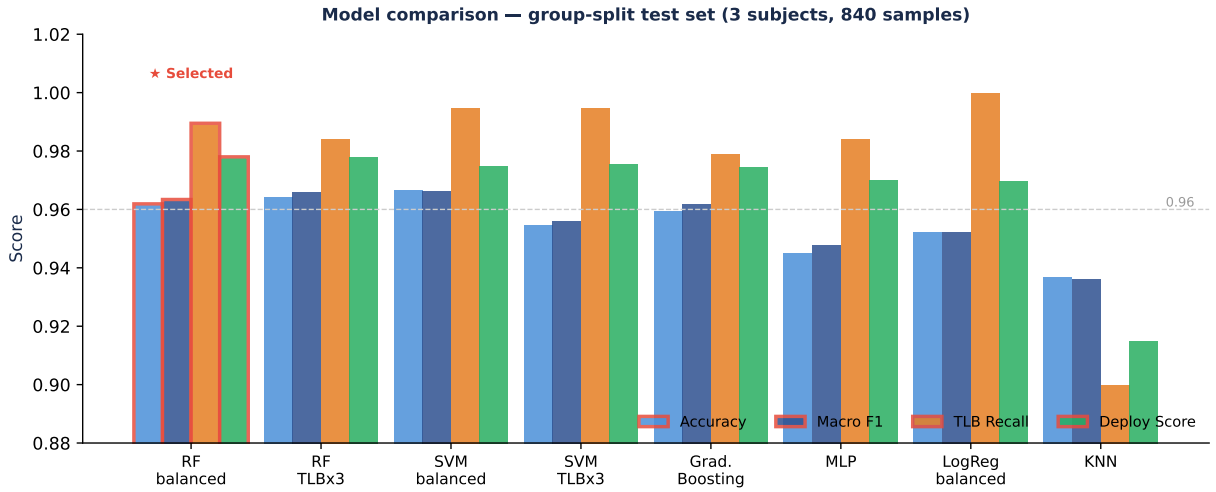

Figure 4: *Model comparison across four metrics on the group-split test set. The selected model (Random Forest balanced, red border) achieves the highest deploy score by balancing overall accuracy, macro F1, and TLB recall simultaneously. KNN is the weakest on TLB recall (0.90); Logistic Regression achieves perfect TLB recall but at the cost of lower TLB precision.*

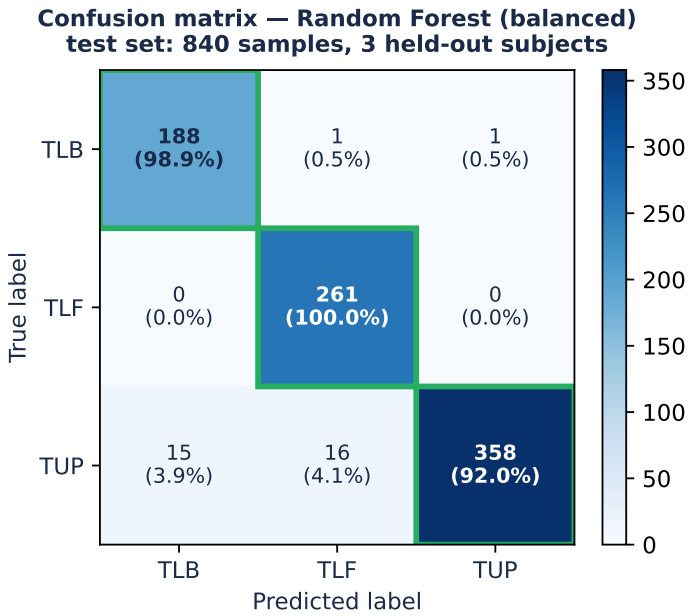

Figure 5: *Confusion matrix of the selected Random Forest (balanced) on the held-out test set (840 samples, 3 subjects). The diagonal is highlighted in green. Nearly all TLB samples are correctly recalled (98.9%); TUP accounts for the majority of off-diagonal errors, where a small fraction of TUP frames is misclassified as TLF or TLB due to subject-specific posture variation.*

8. Real-Time Inference Pipeline

The real-time module integrates OpenCV video capture, MediaPipe Pose estimation, per-subject calibration, and the hybrid classifier into a continuous loop. An iPhone connected as an external camera provides the video feed at 1280×720 pixels.

8.1 Calibration Phase

Before classification begins, the system enters a **calibration phase** of 30 candidate frames (`CALIBRATION_FRAMES = 30`). The user is asked to sit in the neutral TUP posture. Valid frames — those in which all required landmarks exceed the confidence threshold — are accumulated. From these frames the system computes the per-feature mean and standard deviation that define the subject's personal baseline, applying the std floor values described in Section 5.3.

8.2 Classification Loop

Each iteration of the main loop performs the following steps:

1. **Frame capture:** one frame is read from the camera via `cv2.VideoCapture`.
2. **Pose estimation:** MediaPipe Pose (`model_complexity=1`, `smooth_landmarks=True`) extracts world and image landmarks.
3. **Z-smoothing:** a rolling buffer of 10 frames averages the z-coordinates to suppress depth noise.
4. **Feature extraction:** `extract_features` computes all 12 features; only the 3 ML and 4 geometric features are forwarded.
5. **z-score normalisation:** features are normalised relative to the calibrated baseline and clamped to $[-4\sigma, +4\sigma]$.
6. **Hybrid classification:** `classify_posture` fuses the ML prediction with the geometric lateral score and applies hysteresis.
7. **Overlay:** the predicted label, confidence, and landmark skeleton are drawn on the frame.
8. **Logging:** each frame's data is stored; at session end a JSON log and a summary PNG chart are saved.

8.3 Key Parameters

Table 7: *Key runtime parameters and their roles.*

Parameter	Value	Purpose
`CONF_THRESHOLD`	0.5	Minimum MediaPipe landmark visibility to accept a frame
`SMOOTH_WINDOW`	10	Temporal rolling window for z-coordinate smoothing
`CALIBRATION_FRAMES`	30	Number of valid TUP frames to build the subject baseline
`DELTA_CLAMP`	4.0 σ	Maximum z-score deviation before clamping
`HYSTERESIS`	0.15	Minimum score margin to trigger a label change
`TUP_FLOOR`	1.0	Additive TUP probability boost to reduce false alarms

9. Discussion and Limitations

9.1 Strengths

- **High accuracy with minimal features:** three features suffice to achieve 96.2% accuracy, making the model lightweight and fast for real-time use.
- **Subject-adaptive normalisation:** per-subject z-scoring removes the need for large cross-subject training data and makes the system robust to body proportion variability.
- **Interpretable hybrid design:** using geometric rules for lateral classes provides transparency

and allows targeted tuning without retraining.
- **No wearables required:** the system operates with a standard RGB camera, making it accessible and non-intrusive.

9.2 Limitations

- **Class imbalance:** TLB represents only ∼9% of the ML training set. Although class weights mitigate this, performance on TLB may degrade for out-of-distribution subjects.
- **MediaPipe dependency:** landmark quality degrades with occlusion, poor lighting, or side-facing camera angles. No fallback is implemented when too many landmarks fall below threshold.
- **Calibration requirement:** the per-subject baseline must be re-established at the start of each session. Deviations from TUP during calibration can degrade classification accuracy.
- **Limited subject diversity:** 13 subjects may not capture the full range of body types and ambient conditions encountered in production.
- **TLL/TLR not learned:** geometric rules are manually tuned and may require adjustment for subjects with structural asymmetries.

10. Conclusions and Future Work

This project demonstrates that reliable real-time upper-body posture classification is achievable with a minimal set of skeletal features and a hybrid ML–geometric strategy. The Random Forest classifier, trained on just three per-subject normalised features, achieves over 96% accuracy and nearly 99% recall on the minority class TLB. The geometric rules for lateral classes complement the learned model effectively, preventing the most common misclassification between lateral and forward lean.

Several directions are open for future work:

- **Unified 5-class model:** training a single model on all five classes, possibly with data augmentation for TLL and TLR, could simplify the pipeline and improve robustness at the decision boundaries.
- **Larger and more diverse dataset:** expanding to more subjects, varying camera angles, and diverse environments would improve generalisation.
- **Audio or haptic feedback:** augmenting the visual overlay with real-time audio cues would make the system more actionable during screen-intensive tasks.
- **Temporal modelling:** replacing rolling-window smoothing with a sequence model (e.g. LSTM or Transformer) could capture postural drift dynamics and enable early warning before a posture becomes extreme.
- **Automatic recalibration:** detecting when the user's TUP reference has shifted and updating the baseline without user intervention would improve long-session reliability.